

LICHTENBURG, South Africa

WMO: 683410

Lat: 26.133S

Lon: 26.164E

Elev: 1485

StdP: 84.71

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.5	-13.0	1.5	12.7	-11.2	1.7	12.3	8.9	16.8	8.1	17.7	1.7	40	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.7	33.2	16.4	31.7	16.2	30.4	15.9	19.9	26.4	19.4	25.6	19.0	24.9	4.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.7	21.9	17.7	15.3	21.6	17.3	14.8	21.2	64.4	26.6	62.5	25.5	61.0	24.7	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.3	6.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-5.0	20.8	2.5	0.5	-6.9	21.2	-8.3	21.5	-9.8	21.8	-11.6	22.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	22.3	21.9	20.4	17.5	14.2	11.6	11.3	14.4	18.2	20.6	21.4	22.6
	DBStd	4.84	2.11	2.00	2.05	2.52	2.48	2.51	2.65	3.42	3.42	3.26	3.22	2.53
	HDD10.0	39	0	0	0	0	3	12	18	6	1	0	0	0
	HDD18.3	801	1	1	6	44	129	202	218	127	44	17	11	2
	CDD10.0	2957	382	332	322	226	133	61	57	142	246	327	341	389
	CDD18.3	678	124	100	69	19	1	0	0	5	39	86	102	132
	CDH23.3	5429	872	618	418	137	36	3	2	97	431	854	874	1085
Wind	CDH26.7	1733	296	175	81	18	1	0	0	6	116	322	303	417
	WSAvg	2.9	2.9	2.7	2.5	2.4	2.2	2.5	2.6	3.3	3.5	3.5	3.6	3.4
	PrecAvg	547	107	85	81	57	12	6	1	5	14	44	87	81
	PrecMax	1018	258	228	216	165	50	43	16	82	76	149	218	209
	PrecMin	231	16	0	7	0	0	0	0	0	0	0	6	12
	PrecStd	169	55	54	50	43	17	11	3	14	19	33	52	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	32.7	30.7	28.7	26.2	23.2	23.1	27.5	31.3	34.3	34.1	35.6
		MCWB	17.9	17.1	16.9	15.8	12.5	10.6	9.7	12.7	13.7	15.6	15.9	16.8
	2%	DB	32.2	30.8	28.9	26.7	24.4	21.3	21.1	25.9	29.6	32.0	32.0	32.7
		MCWB	17.5	17.4	17.0	15.6	12.2	9.8	9.2	12.0	13.2	14.9	15.7	17.1
	5%	DB	30.1	29.0	27.2	25.0	22.9	19.9	19.5	24.1	27.9	30.4	30.2	30.8
		MCWB	17.5	18.0	17.2	15.3	11.9	9.5	8.9	11.2	12.6	14.4	15.4	16.8
	10%	DB	28.1	27.2	25.8	23.3	21.0	18.1	17.7	22.0	25.9	28.5	28.4	28.9
		MCWB	17.9	18.2	16.9	14.9	11.4	8.8	8.2	10.0	12.0	13.9	15.5	16.7
	0.4%	WB	20.7	20.7	19.7	18.4	15.0	12.0	11.5	14.0	16.7	17.6	19.1	19.7
		MCDB	28.2	27.4	26.1	24.5	22.8	19.9	19.6	24.7	24.0	28.6	27.1	28.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.0	20.0	19.1	17.5	13.8	11.1	10.5	13.1	15.0	16.5	18.3	19.1
		MCDB	26.2	25.9	25.4	23.4	21.0	19.1	18.6	23.2	25.1	26.6	26.2	27.3
	5%	WB	19.4	19.5	18.5	16.8	13.0	10.3	9.7	12.1	13.9	15.8	17.6	18.5
		MCDB	25.2	25.0	24.2	22.3	20.5	18.0	17.7	20.8	24.5	25.7	25.4	26.3
	10%	WB	19.0	19.0	17.9	16.0	12.2	9.4	8.9	11.2	13.1	15.2	16.9	18.1
		MCDB	24.4	24.0	22.9	21.1	19.3	16.6	16.7	19.5	22.9	24.7	24.5	25.4
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.2	10.8	11.7	14.8	15.4	14.8	15.2	15.1	13.9	12.8	11.7
		MCDBR	14.5	14.0	13.4	14.0	16.8	16.7	16.1	16.6	16.9	16.3	15.2	15.0
		MCWBR	4.7	4.9	5.0	5.2	7.3	7.8	7.5	7.6	7.0	6.3	5.2	5.0
	5% WB	MCDBR	9.8	9.9	10.5	10.9	14.3	14.9	14.2	14.6	14.2	13.1	12.1	11.4
		MCWBR	3.8	4.1	4.4	4.8	6.5	7.4	7.1	7.1	6.2	5.3	4.4	4.3
Clear-Sky Solar Irradiance	taub		0.362	0.367	0.352	0.316	0.279	0.264	0.263	0.316	0.360	0.386	0.340	0.348
	taud		2.397	2.378	2.405	2.475	2.553	2.565	2.541	2.379	2.263	2.229	2.414	2.425
	Ebn at Noon		981	959	944	936	933	930	943	920	921	930	996	996
	Edh at Noon		128	127	117	100	83	78	83	108	133	146	125	125
All-Sky Solar Radiation	RadAvg		6.93	6.41	5.73	4.85	4.40	3.96	4.31	5.14	6.21	6.75	7.13	7.22
	RadStd		0.71	0.59	0.49	0.41	0.19	0.17	0.11	0.24	0.25	0.41	0.57	0.43

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page